🔌 EduTrack — API Documentation

Academic Management Platform | VTU Internship 2026 — Team 15 | April 2026

Overview

All APIs return JSON responses. Every response includes a `success` boolean field.

Base URL (Local): `http://localhost:8080`

Base URL (Production): `https://edutrack-backend-o1ym.onrender.com`

1. Authentication APIs

1.1 Admin Login

METHOD	POST
ENDPOINT	`/api/auth/login/admin`
DESCRIPTION	Authenticate admin user

Request Body:

```
{
  "email": "admin@edutrack.com",
  "password": "Admin@123"
}
```

Success Response:

```
{
  "success": true,
  "adminId": 1,
  "name": "Admin",
  "email": "admin@edutrack.com",
  "role": "admin",
```

```
  "message": "Login successful"
}
```

1.2 Student Login

METHOD	**POST**
ENDPOINT	`/api/auth/login/student`
DESCRIPTION	Authenticate student user

```
{
  "email": "student@college.edu",
  "password": "Test@123"
}
```

Success Response:

```
{
  "success": true,
  "studentId": 1,
  "name": "Arjun Kumar",
  "email": "student@college.edu",
  "department": "Computer Science and Engineering",
  "year": "3rd Year",
  "semester": "Sem 5",
  "usn": "1VT22CS001",
  "phone": "9876543210",
  "role": "student",
  "message": "Login successful"
}
```

1.3 Teacher Login

METHOD	**POST**
ENDPOINT	`/api/auth/login/teacher`
DESCRIPTION	Authenticate teacher user

```
{
  "email": "teacher@college.edu",
```

```
  "password": "Test@123"
}
```

1.4 Student Registration

METHOD	**POST**
ENDPOINT	`/api/auth/register/student`
DESCRIPTION	Self-registration for students (blocked if registrationOpen=false)

```
{
  "name": "Arjun Kumar",
  "email": "arjun@college.edu",
  "password": "Test@123",
  "department": "Computer Science and Engineering",
  "year": "1st Year",
  "semester": "Sem 1",
  "phone": "9876543210",
  "usn": "1VT25CS001"
}
```

1.5 Admin Add Student

METHOD	**POST**
ENDPOINT	`/api/auth/admin/add-student`
DESCRIPTION	Admin adds student (bypasses registration gate)

Request Body: Same as student registration

1.6 Admin Add Teacher

METHOD	**POST**
ENDPOINT	`/api/auth/admin/add-teacher`
DESCRIPTION	Admin adds teacher

```
{
  "name": "Prof. Sharma",
  "email": "sharma@college.edu",
```

```
  "password": "Test@123",
  "department": "Computer Science and Engineering",
  "phone": "9876543210",
  "employeeId": "EMP001"
}
```

1.7 Change Admin Password

METHOD	**POST**
ENDPOINT	`/api/auth/change-password/admin`
DESCRIPTION	Change admin password (requires current password)

```
{
  "id": 1,
  "currentPassword": "Admin@123",
  "newPassword": "NewPass@123"
}
```

1.8 Change Student Password

METHOD	**POST**
ENDPOINT	`/api/auth/change-password/student`
DESCRIPTION	Change student password

```
{
  "id": 1,
  "currentPassword": "Test@123",
  "newPassword": "NewPass@123"
}
```

1.9 Change Teacher Password

METHOD	**POST**
ENDPOINT	`/api/auth/change-password/teacher`
DESCRIPTION	Change teacher password

Request Body: Same as change student password

2. OTP / Password Reset APIs

2.1 Send OTP (Forgot Password)

METHOD	POST
ENDPOINT	`/api/otp/forgot-password`
DESCRIPTION	Send OTP to email for password reset. Validates email exists for given role.

```
{
  "email": "student@college.edu",
  "role": "student"
}
```

role can be: student, teacher, admin

2.2 Verify OTP

METHOD	POST
ENDPOINT	`/api/otp/verify`
DESCRIPTION	Verify OTP code. OTP expires in 10 minutes and is single-use.

```
{
  "email": "student@college.edu",
  "otp": "123456"
}
```

2.3 Reset Student Password

METHOD	POST
ENDPOINT	`/api/auth/reset-password/student`
DESCRIPTION	Reset student password after OTP verification

```
{
  "email": "student@college.edu",
  "otp": "123456",
  "newPassword": "NewPass@123"
}
```

2.4 Reset Teacher Password

METHOD	POST
ENDPOINT	`/api/auth/reset-password/teacher`
DESCRIPTION	Same as reset student password

2.5 Reset Admin Password

METHOD	POST
ENDPOINT	`/api/auth/reset-password/admin`
DESCRIPTION	Same as reset student password

3. Student APIs

METHOD	ENDPOINT	DESCRIPTION
GET	`/api/students`	Get all students
GET	`/api/students/{id}`	Get student by ID
PUT	`/api/students/{id}`	Update student profile
DELETE	`/api/students/{id}`	Delete student (cascades attendance, marks, submissions)
PATCH	`/api/students/{id}/status`	Update student status (Active/Suspended/Graduated)
POST	`/api/students/{id}/reset-password`	Admin force reset student password

Update Student Profile – Request Body:

```
{
  "name": "Updated Name",
  "phone": "9999999999"
}
```

4. Teacher APIs

METHOD	ENDPOINT	DESCRIPTION
GET	/api/teachers	Get all teachers
GET	/api/teachers/{id}	Get teacher by ID
PUT	/api/teachers/{id}	Update teacher profile
DELETE	/api/teachers/{id}	Delete teacher (cascades courses, attendance, marks, etc.)
PATCH	/api/teachers/{id}/status	Update teacher status (Active/Suspended)

Update Teacher – Request Body:

```
{
  "name": "Updated Name",
  "phone": "9999999999",
  "employeeId": "EMP002"
}
```

5. Course APIs

METHOD	ENDPOINT	DESCRIPTION
GET	/api/courses	Get all courses
GET	/api/courses/{id}	Get course by ID
GET	/api/courses/teacher/{teacherId}	Get courses by teacher
GET	/api/courses/department/{department}	Get courses by department
POST	/api/courses	Create course

PUT	/api/courses/{id}	Update course
DELETE	/api/courses/{id}	Delete course (cascades attendance, marks, assignments, materials)

Create/Update Course – Request Body:

```
{
  "courseName": "Data Structures",
  "credits": 4,
  "teacher": { "teacherId": 1 }
}
```

6. Attendance APIs

METHOD	ENDPOINT	DESCRIPTION
GET	/api/attendance/student/{studentId}	Get attendance by student
GET	/api/attendance/course/{courseId}	Get attendance by course
GET	/api/attendance/course/{courseId}/date/{date}	Get attendance by course and date (YYYY-MM-DD)
GET	/api/attendance/course/{courseId}/stats	Get attendance stats (percentages per student)
POST	/api/attendance	Mark attendance (upsert)
PUT	/api/attendance/{id}	Update attendance record

Mark Attendance – Request Body:

```
{
  "student": { "studentId": 1 },
  "course": { "courseId": 1 },
  "attendanceDate": "2026-04-21",
  "status": "Present"
}
```

7. Marks APIs

METHOD	ENDPOINT	DESCRIPTION
GET	/api/marks/student/{studentId}	Get marks by student
GET	/api/marks/course/{courseId}	Get marks by course
POST	/api/marks	Enter marks (auto-calculates grade)
PUT	/api/marks/{id}	Update marks
DELETE	/api/marks/{id}	Delete marks

Enter/Update Marks – Request Body:

```
{
  "student": { "studentId": 1 },
  "course": { "courseId": 1 },
  "marks": 85
}
```

8. Assignment APIs

METHOD	ENDPOINT	DESCRIPTION
GET	/api/assignments	Get all assignments
GET	/api/assignments/course/{courseId}	Get assignments by course
POST	/api/assignments	Create assignment
PUT	/api/assignments/{id}	Update assignment

DELETE	/api/assignments/{id}	Delete assignment

Create Assignment – Request Body:

```
{
  "course": { "courseId": 1 },
  "title": "Linked List Implementation",
  "description": "Implement singly and doubly linked lists",
  "dueDate": "2026-05-01"
}
```

9. Course Materials APIs

9.1 Get Materials by Course

METHOD	GET
ENDPOINT	/api/materials/course/{courseId}

9.2 Upload Material (File)

METHOD	POST
ENDPOINT	/api/materials/upload
REQUEST	multipart/form-data

PARAMETER	TYPE	DESCRIPTION
courseId	Integer	Course ID
teacherId	Integer	Teacher ID
title	String	Material title
description	String	Optional description
file	File	File to upload

9.3 Add Material (Link)

METHOD	POST

ENDPOINT	`/api/materials/link`

```
{
  "courseId": "1",
  "teacherId": "1",
  "title": "Reference Notes",
  "description": "Optional",
  "url": "https://example.com/notes"
}
```

9.4 Download Material

METHOD	GET
ENDPOINT	`/api/materials/download/{id}`

9.5 Delete Material

METHOD	DELETE
ENDPOINT	`/api/materials/{id}`

10. Submission APIs

10.1 Submit Assignment (File Upload)

METHOD	POST
ENDPOINT	`/api/submissions/upload`
REQUEST	multipart/form-data

PARAMETER	TYPE	DESCRIPTION
assignmentId	Integer	Assignment ID
studentId	Integer	Student ID
file	File	Submission file

10.2 Get Submissions by Assignment

METHOD	GET
ENDPOINT	/api/submissions/assignment/{assignmentId}

10.3 Get Submissions by Student

METHOD	GET
ENDPOINT	/api/submissions/student/{studentId}

10.4 Check Submission Status

METHOD	GET
ENDPOINT	/api/submissions/check?assignmentId={id}&studentId={id}

Response:

```
{
  "submitted": true,
  "submissionId": 1,
  "fileName": "assignment.pdf",
  "submittedAt": "2026-04-21T10:30:00",
  "grade": "A+",
  "comment": "Good work"
}
```

10.5 Grade Submission

METHOD	PUT
ENDPOINT	/api/submissions/{id}/grade

```
{
  "grade": "A+",
  "comment": "Excellent work"
}
```

10.6 Download Submission

METHOD	GET
ENDPOINT	/api/submissions/download/{submissionId}

11. Health API

11.1 Database Health Check

METHOD	GET
ENDPOINT	/api/health/db

Response:

```
{ "status": "Connected" }
```

12. System Settings APIs

12.1 Get System Settings

METHOD	GET
ENDPOINT	/api/system/settings

```
{
  "id": 1,
  "activeYear": "2025-26",
  "activeSemester": "Even",
  "maintenanceMode": false,
  "registrationOpen": true
}
```

12.2 Update System Settings

METHOD	PUT
ENDPOINT	/api/system/settings

```
{
  "maintenanceMode": true,
  "registrationOpen": false,
  "activeYear": "2025-26",
  "activeSemester": "Even"
}
```

13. Bulk Import APIs

13.1 Bulk Import Students

METHOD	POST
ENDPOINT	`/api/bulk/students`
REQUEST	multipart/form-data with CSV file

CSV Format:

```
name,email,password,department,year,semester,phone,usn
Arjun Kumar,arjun@college.edu,Test@123,Computer Science and Engineering,3rd
```

13.2 Bulk Import Teachers

METHOD	POST
ENDPOINT	`/api/bulk/teachers`
REQUEST	multipart/form-data with CSV file

CSV Format:

```
name,email,password,department,phone,employeeId
Prof. Sharma,sharma@college.edu,Test@123,Computer Science and Engineering,98
```

13.3 Bulk Import Attendance

METHOD	POST

ENDPOINT	`/api/bulk/attendance?courseId={id}&date={date}`
REQUEST	multipart/form-data with CSV file

Query Parameters:

`courseId` – ID of the course

`date` – Date in `YYYY-MM-DD` format

CSV Format:

```
usn,status
1VT22CS001,Present
1VT22CS002,Absent
1VT22CS003,Present
```

Response: `{"success": true, "imported": 3, "errors": []}`

Upsert: updates existing record if same student+course+date already exists

13.4 Bulk Import Marks

METHOD	**POST**
ENDPOINT	`/api/bulk/marks?courseId={id}`
REQUEST	multipart/form-data with CSV file

CSV Format:

```
usn,marks
1VT22CS001,85
1VT22CS002,72
1VT22CS003,91
```

Grade is auto-calculated. Updates existing record if already present.

API Summary

MODULE	TOTAL ENDPOINTS
Authentication	9

MODULE	TOTAL ENDPOINTS
OTP / Password Reset	5
Students	6
Teachers	5
Courses	7
Attendance	6
Marks	5
Assignments	5
Course Materials	5
Submissions	6
Health	1
System Settings	2
Bulk Import	4
Total	**65**